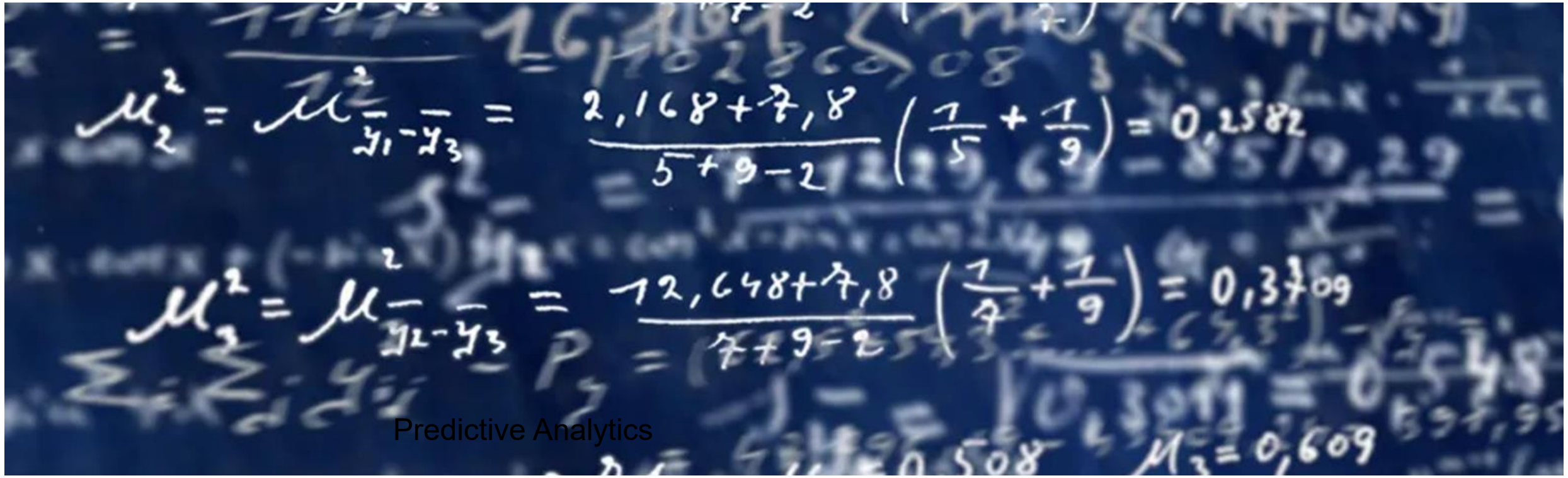

Predictive Analytics · Lecture 06

INTRODUCTION TO ARIMA MODELS

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AGENDA

- The story so far
- ARIMA models – what are they?
- Tools of the trade:
    1. The backshift operator
    2. The autocorrelation function
    3. The partial autocorrelation function
- Fooling around in Excel

THE STORY SO FAR

THE PREDICTION CHALLENGE

We want to forecast $y_{t+1}, y_{t+2}, \ldots, y_{t+h}$ at time t

Suppose

$$y_{t+1} = f(x_t) + \varepsilon_{t+1}$$

where

- y is the variable of interest
- x is the predictor set
- ε is random noise

So many questions…

- f(•) is the Data Generating Process, DGP
  - Does it include **unobserved** or **unknown** variables?
  - Is it based on information that can not be acquired at all?
  - And most of all: is it a **stable process**?
- What is in x ?
  - History of y
  - Independent predictors
- Which methodology?
  - Parametric approach (econometrics)
  - Nonparametric approach (ML)
- What is the quality of the forecast?
  - How to define "quality"
  - Unbiased vs.
    "closer to true value, but always a little off-target"

ARIMA MODELS

WHAT ARE THEY?

ARIMA: AUTOREGRESSIVE INTEGRATED MOVING AVERAGE

AR MA

$$\Delta_d x_t = c + \phi_1 \Delta_d x_{t-1} + \ldots + \phi_p \Delta_d x_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

or…

$$y_t = c + \phi_1 y_{t-1} + \ldots + \phi_p y_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

where $y_t = \Delta_d x_t$

Autoregressive component (AR) component

- The current value depends on its past values.
- Regression on lagged values of the variable itself.
- Order denoted by **p** - the number of time lags of the variable included in the model.

Integration (I) component

- Involves differencing the time series to achieve *stationarity*.
- Degree of differencing denoted by **d** - the number of times differencing is applied.
- Actual model variable is the **d'th** difference between consecutive values.

Moving Average component (MA)

- The current value of the model variable depends on current and past values of the error terms.
- Order denoted by **q** - the number of lagged errors included.

ARIMA NOTATION AND EXAMPLES

$$\Delta_d x_t = c + \phi_1 \Delta_d x_t + \ldots + \phi_p \Delta_d x_{t-p} + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q} + \varepsilon_t$$

- ARIMA(p, d, q) represents the order of each of the three components.
- Seasonal ARIMA (SARIMA): ARIMA(p, d, q)(P, D, Q)m.

- Examples
  - ARIMA(0, 0, 0): *White Noise* $x_t = \varepsilon_t$
  - ARIMA(0, 1, 0): *Random Walk* $\Delta x_t = \varepsilon_t \iff x_t - x_{t-1} = \varepsilon_t \iff x_t = x_{t-1} + \varepsilon_t$
  - ARIMA(p, 0, 0): *Autoregressive AR(p)* $x_t = c + \phi_1 x_{t-1} + \ldots + \phi_p x_{t-p} + \varepsilon_t$
  - ARIMA(0, 0, q): *Moving Average MA(q)* $x_t = c + \varepsilon_t + \theta_1 \varepsilon_{t-1} + \ldots + \theta_q \varepsilon_{t-q}$
  - ARIMA(0, 1, 1): *Basic Exponential Smoothing* $x_t = x_{t-1} + \varepsilon_t + \theta_1 \varepsilon_{t-1}$

WHY ARE ARIMA MODELS IMPORTANT?

- Among the most widely used techniques for time series forecasting.
- Effective for univariate time series with trends and autocorrelation.
- Provide a way to model the internal dynamics of a time series.
- Complementary to exponential smoothing methods (which focus on trend and seasonality):

$$y_t = T_t + S_t + R_t$$

EXAMPLES INCLUDE:

- *Macroeconomic Forecasting:*
  Predicting GDP growth, inflation rates, unemployment figures.
- *Financial Market Analysis:*
  Forecasting stock prices, exchange rates.
- *Sales and Demand Forecasting:*
  Predicting product demand for inventory management and business planning.
- *Energy and Climate Forecasting:*
  Predicting electricity consumption, temperature trends.

LIMITATIONS

- Require a **long history** of data for reliable model fitting.
- Assumes **stability of the Data Generating Process** i.e. assumes that past patterns will continue into the future.
- **Can be challenging** to correctly identify the order (p, d, q).
- Basic ARIMA **does not directly incorporate external (exogenous) variables** (for this, consider ARIMAX).

TOOLS OF THE TRADE

THE BACKSHIFT OPERATOR B - AKA THE LAG OPERATOR L

- The backshift operator **B** shifts the time index backward by one period:

$$By_t = y_{t-1}$$

- Conventional algebraic operations apply. Here are some useful examples:

$$B^2 y_t = B(By_t) = By_{t-1} = y_{t-2}$$

$$B^n y_t = y_{t-n}$$

$$(1 - B)\, y_t = y_t - y_{t-1} = \Delta y_t$$

THE BACKSHIFT OPERATOR ALLOWS FOR CONCISE REPRESENTATION OF ARIMA MODELS

$$\frac{1}{1-x} = 1 + x + x^2 + x^3 + \dots$$

$$B \quad B^2$$

$$(1 - \phi_1 B + \dots + \phi_p B^p)(1 - B)^d x_t = (1 + \theta_1 B + \dots + \theta_q B^q)\,\varepsilon_t$$

$$\Delta x_t = (1 - B)\, x_t$$

Random walk process:

$$(1 - B)\, x_t = \varepsilon_t \Leftrightarrow x_t = \frac{1}{1-B}\,\varepsilon_t = \varepsilon_t + \varepsilon_{t-1} + \varepsilon_{t-2} + \varepsilon_{t-3} + \dots$$

THE AUTOCORRELATION FUNCTION (ACF)

$$r_k = \frac{\sum_{t=k+1}^{T} (y_t - \bar{y})(y_{t-k} - \bar{y})}{\sum_{t=1}^{T} (y_t - \bar{y})^2}$$

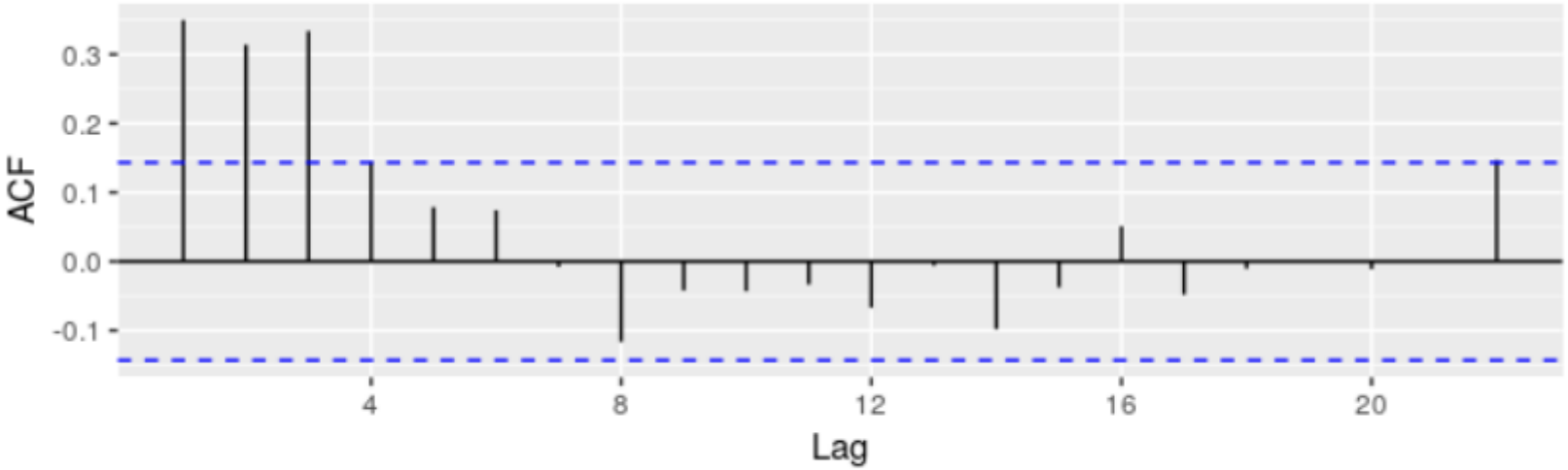

Figure 8.9: ACF of quarterly percentage change in US consumption.

THE PARTIAL AUTOCORRELATION FUNCTION (PACF)

- Measures the correlation between a time series and its lagged values after removing the influence of the intermediate lags•
- Helps in identifying the order of the AR component (p)

ACF: $x_t \leftrightarrow x_{t-k}$

PACF: $x_t \leftrightarrow x_{t-k}$

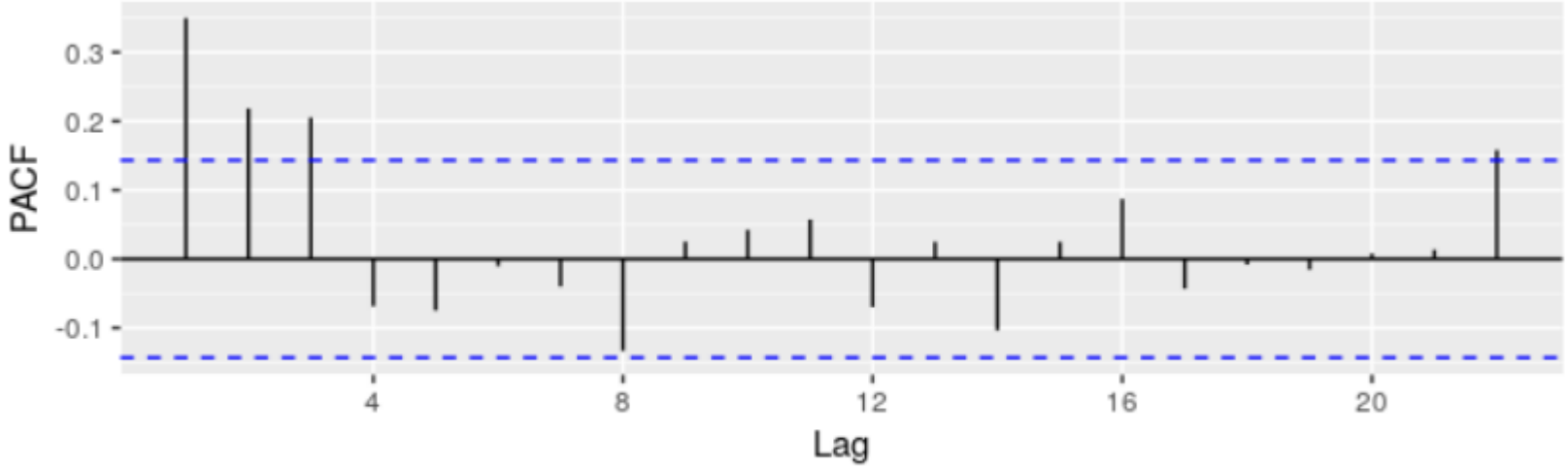

Figure 8.10: PACF of quarterly percentage change in US consumption.

STATIONARITY

- A stationary time series has statistical properties (mean, variance, autocovariance) that remain constant over time
- Trends and seasonality indicate non-stationarity
- ARIMA modeling requires the time series to be stationary
- Why stationarity? Many statistical properties of time series models rely on this assumption.

How?

- **Differencing** between consecutive observations: 1 – B
- If that is not enough, then take second difference
- …and so on
- **Seasonal differencing** can be used to remove seasonal patterns1 ....

FOOLING AROUND IN EXCEL

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